DIU SELF REGISTRATION SYSTEM

By
Mehedi Hasan Munna
221-15-5148

Nime Ullaha
221-15-5515

FINAL YEAR DESIGN PROJECT REPORT

This Report Presented in Partial Fulfillment of the Requirements for the **Degree of Bachelor of Science in Computer Science and Engineering**

Supervised by

Fatema Tuj Johora
Assistant Professor
Department of Computer Science and Engineering Daffodil International University

Co-Supervised by

Most. Hasna Hena
Assistant Professor
Department of Computer Science and Engineering Daffodil International University

DAFFODIL INTERNATIONAL UNIVERSITY
Dhaka, Bangladesh

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APPROVAL

This Project titled "**DIU SELF REGISTRATION SYSTEM**," submitted by **Mehedi Hasan Munna** and **Nime Ullaha** to the Department of Computer Science and Engineering, Daffodil International University, has been accepted as satisfactory for the partial fulfillment of the requirements for the degree of B.Sc. in Computer Science and Engineering and approved as to its style and contents. The presentation was held on 10-03-2025.

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Md. Ridoy Sarkar
Internal Examiner 2
Lecturer, Department of CSE, FSIT
Daffodil International University

_____

Name
External Examiner
Designation, Department of CSE, FSIT
Daffodil International University

DECLARATION

We hereby declare that this project has been done by us under the supervision of **Fatema Tuj Johora**, **Assistant Professor**, Department of Computer Science and Engineering, Daffodil International University. We also declare that neither this project nor any part of this project has been submitted elsewhere for the award of any degree or diploma.

Supervised by:

Fatema Tuj Johora
Assistant Professor
Department of Computer Science and
Engineering Daffodil International University

Co-Supervised by:

Most. Hasna Hena
Assistant Professor
Department of Computer Science and
Engineering Daffodil International University

Submitted by:

Mehedi Hasan Munna
Student ID : 221-15-5148
Department of Computer Science and
Engineering Daffodil International University

Nime Ullaha
Student ID : 221-15-5515
Department of Computer Science and
Engineering Daffodil University

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ABSTRACT

DIU Self Registration System is a client-server application that combines the benefits of online and offline applications. In the past, students at DIU relied on their academic advisers to handle course registration manually — a process that not only causes delays and miscommunication among students and tutors, but also results in incorrectly selected courses and a heavy administrative burden for staff. This project solves these problems by providing a self-service registration system that allows students to search for courses, check prerequisites, identify scheduling conflicts and register in real time. The app is developed with Next.js, which allows a Node.js backend and a React frontend without any barriers, providing a fluid data flow and a responsive user experience. Tailwind CSS is used to create a clean, user-friendly and mobile-responsive UI and MongoDB is used as a flexible, scalable database to store information about students, course offerings, registrations and user roles.Further, authenticated sessions and user-type based permissions are implemented so that students, Teachers and admin have suitably varying permissions and tool sets to perform their roles. The DIU Self Registration System streamlines the process and reduces the paperwork for the staff, minimizes errors, allows faster processing of the registration forms and makes the whole process more transparent. As far as convenience is concerned, students can access the platform to have more control and a clear view on their course choices. From the institutional perspective it is aligned with DIU's vision for the future, which is to transform academic services through digitalization. Ultimately, this project represents a pragmatic, full-stack real-world solution to a practical academic management challenge – bringing benefits in terms of efficiency, user satisfaction and institution modernization.

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Chapter 1

Introduction

This chapter describes the DIU Self Registration System and presents the background, rationale, goals, approach, anticipated results, and the organization of the full report.

1.1 Introduction

Enrolling in courses is a pivotal academic affair in any university as it directly impacts students' academic life and graduation in time. At Daffodil International University (DIU), the system used to be that student advisors take course registration on students' behalf. Although the approach has proved effective over the years, it often leads to a slow paced process, bureaucratic bottlenecks and dependence on students. Course registration is an important academic activity in any university as it determines students' academic life and graduation in time. At Daffodil International University (DIU), the procedure has been that student advisors register courses for students. Although that approach has served over many years, it frequently resulted to tedious procedures, administrative bottlenecks and relied-on-the-students processes.

At most universities, the student course registration system has to be accurate, timely and easy to use. Traditionally, DIU has relied on semi-manual, or at best, partially automated systems for student registration, which are prone to inefficiencies such as long queues, data entry errors and delays in course enrollment. With the exponential growth in the number of students and the increasing import of technology, a user (including a remote security manager) will definitely want a simpler, more secure and a more self-service registration system.

The aim of this system is the reduction of reliance on course advisors, along with the errors, double bookings and user frustrations and to have a better registration experience. Also the system enforces real-time validation of prerequisites, credit limits, time collisions, etc, which ensures that the registration is academically sound and in compliance with the school policies.This report presents the development process of Daffodil International University (DIU) Self Registration System including planning, designing, implementing and testing and discusses the importance of the project in enhancing automation and digitization within the service of DIU academic.

1.2 Motivation

The current marquee registration system at DIU is at its core under the control of the academic advisors, the process while necessary for providing essential guidance contributes to make the student experience is the introduction of unnecessary friction and

inefficiency. This reliance creates a mandatory and frequently restrictive, bottleneck that students must get through if they are to do their vital registration business.One of the main difficulties is that the registration requirement is synchronous students must locate an advisor's availability within what are usually limited and in high demand registration slots. This directly contributes to delays, especially in the rush of registration when advisors are booked fully. The delay these causes not only inconvenience students, but also add to their stress and can cause feelings of bitterness towards the bureaucracy. Centralization of control over the process of approval, brings to the fore any temporary unavailability of an advisor seem to be a single point of failure in the system an advisor may be busy with prior commitments, or face unexpected circumstances, or simply overwhelmed with too many simultaneous student requests. The bottleneck interrupts the smooth flow of registration, which can have an adverse affect on students being able to register in a timely manner and on the institution's ability to make use of its resources.Taking all of these inherent challenges into account long queues for users, system bottlenecks and reliance on synchronous human-based operations it seems to us that there is a surprisingly large and compelling opportunity to develop a more efficient, automated and student-centered registration system. This work is motivated by the aspiration to eliminate these systemic inefficiencies and ease course registration for students of DIU.

1.3 Objectives

The general aim of this project is the process reform, i.e., modernization of DIU course registration process and introduction of a digital, advisor-less system for the registration. The development activities for this project are defined by the following specific, measurable success criteria:

- The main goal is to design and implement a web-based system that supports secure, efficient, and easy self-registration of students for their courses.
- Minimize the manual work for the academic advisers and the administrative staff.
- Students can register for classes from the comfort of their own home via a web portal.
- Improve the student experience by providing a quicker, more dependable, and open registration process.

1.4 Methodology

In this paper, a systematic procedure for a university's self-course registration system is presented which starts from student end while they login into the course registration portal with their institutional email and password. At first the system checks if the period of registration is open, if not it will wait. When registration becomes active, the system determines if the student has completed the requisite teaching evaluations; if not, the student is notified to complete them before moving forward. After he is cleared, the student

on your system chooses do you want to “Enroll Course”, the student selects his desired course and the system will look any prerequisite issue. If there are prerequisites, they must add those courses first before moving on; if not, they may continue to pick a section. The student adds the selected course to their list, completes the registration and at the end of the process receives a notification of successful enrolment.

1.5 Project Outcome

The DIU Self-Registration System, available for use by students now, effortlessly automates course registration from a previously manual, advisor-dependent process. At the heart is a responsive Web Portal (developed using Next.js, Tailwind CSS and ShadCN) that communicates with an Automated Compliance Engine (ACE) which rule-wise enforces every academic rule including pre req checking, credit hour limits, in real-time. This process leads directly to major efficiency gains in operations, most notably a 60 percent reduction in manual administrative burdens on academic advisors. Rendering synchronous advisor approval unnecessary, breaking registration bottlenecks, students can now be instantaneously enrolled from anywhere, at any time, which enriches the user experience drastically. The system also has a far higher level of data integrity (due to automatic rules compliance) and it provides a strong, modular base for the future digitization of DIU academics.

1.6 Organization of the Report

This report is organized as six chapters. The section 2 describes background and related work to the system. Research procedures and system architecture are described in Chapter 3. Implementation, Evaluation and Results are in Chapter 4. Chapter 5 addresses engineering standards, social implications and management issues of the project. Finally, Chapter 6 summarizes the report, describes the limitations and provides recommendations for future work.

Chapter 2
Background

A comprehensive background review with past work, review on related systems, determination of constraints of the existing system and motivation for developing the proposed registration system.

2.1 Introduction

In this chapter, the current university registration systems with their shortcomings are discussed and the need for the development of the DIU Self-Registration System is rationalized. In addition, the proposed system will discuss to address using current environment issues and its impact on using various technologies that could satisfy the users better than before.

Digital course registration systems are now imperative to contemporary university education, allowing students to participate in their academic life with more autonomy and efficacy. Though a few universities of Bangladesh including IUB, AIUB and NSU have implemented system for online registration but still this kind of systems are not free from limitations they have very poor mobile responsiveness, the real-time feedback is also not up to the mark and there are some usability related issues which are really daunting when it comes to consider overall student experience. To fill these gaps DIU Self-Registration System is based on a lightweight, contemporary and user-friendly architecture leveraging the power of Next.js, MongoDB and Tailwind CSS. This system will help to achieve a more user friendly, flexible and faster course registration procedure for the DIU students keeping pace with contemporary technology and student requirement.

2.2 Literature Review

With the help of a system based on a well-structured and systematically organized online system, students can get registered swiftly. The registration is conducted by the University's Registrar's office, through the Student Portal, RDS (Registration and Data System). Using institutional credentials, students log in to see which courses they are eligible to take, and to register during predetermined registration windows. The system, while robust in terms of functionality, is rigid and inflexible; it is continually plagued by server congestion during peak registration hours, and requires manual intervention in case of scheduling conflicts between courses, greatly inconveniencing students and increasing the wait time [1]. A portal-based self-registration system at United International University enables students to choose courses and sections in line with the academic calendar. The student portal is among other academic systems—such as the eLearning Management System (eLMS) that keep course registration and teaching delivery in sync. Although system enforces administrative uniformity and academic coherence, it is rather reluctant to modern usability features (like real-time notification, conflict visualization, and mobile first design) that are becoming more and more relevant for student centered systems [2]. East West University uses an online

student portal for semester-wise registration and add-drop to facilitate the students. Students can log into the system using their institutional credentials and can register for courses within their department, by academic standing, and by predefined credit limits. The paperwork and manual office work were reduced significantly by the system. But it is mostly a standard portal design with little real-time feedback, no enhanced user interface and no mobile solutions [3]. The BRAC University has a well-organized self-registration system which is overseen by the Office of the Registrar. Students engage in self-registration during their respective time slots in registration rounds based on their academic standing and number of credits earned. It is governed by academic rules and credit limitations that make for a controlled and orderly enrollment via the on-line site. While this tightly controlled system dramatically reduces the work load for administration, it is designed for control of process rather than caring about users, offering very little humanization or interactive feedback during registration [4]. AIUB offers an online course registration system which allows students to log in, pick courses and add/drop with little advisor intervention. The system automates a number of critical registration processes and enforces academic rules. Although the system appears to be stable and fully functional, the lack of standard modern user interface components, the absence of mobile friendliness, and the lack of real-time notification to inform students about registration status limits the usability and user satisfaction (perceived usefulness and perceived ease of use) of the system [5].IUB utilizes a comprehensive Enterprise Resource Planning (ERP) system for student course registration. Students can register by credits, check for conflicts, and petition for credit all via the online registration site. The system is highly functional and allows for efficient administration, but does not specifically focus on mobile optimization or user-centered design, which are increasingly considered best practices for contemporary academic applications [6].

2.2.1 Similar Applications

In this chapter, we present the like course registration systems of the other universities and in brief compare their salient features with the proposed DIU Self-Registration SYSTEM.

1. North South University : NSU operates an online advising and registration system RDS, through which students make course selections during assigned time sessions. It operates on a pure first come, first served basis,and is so competitive. Students can browse the courses that are on offer, check the prerequisites, avoid schedule clashes and book their seats straight away. It works, but because of intense competition in rush time, it often leads to anxiety among students[1].

2. United International University : UIU has an online student portal that facilitates course pre advising, advising, payment and add/drop transactions through digital means. The portal displays information such as (sub)available sections of the course, time schedules, course capacity and warnings for (missing) prerequisites. Students are enabled to finish both pre-advising and formal advising in the system. UIU's system is popular but still it heavily

relies on first come first serve registration[2].

3. East West University : EWU has implemented an online advising portal for use by pre-advisors, advisors and students in the add/drop process. They pick their courses and sections through the portal. The screen also lists whether seats are available, time conflicts and prerequisites have been met. Similar to UIU, the system is a racing FCFS model that forces students to rush when the system opens[3].

4. Brac University : BRACU employs the Student Portal as well as buX for planning and registering for courses. The students are provided with very specific individual time slots according to how many credits they have earned. During their slot, they log in and pick sections. This method minimizes server competition and allows the server to more efficiently handle users. The system also incorporates the whole class permission application process and verifies that payment is made prior to registration confirmation[4].

5.American University of Bangladesh : Advisor permission to register system, VUES, is in place at AIUB and advisors own course and section assignment decisions. Students go through a step-by-step procedure: consult the schedule, pre-register via the AIUB Portal, see advisors for problems, and pay for final confirmation. Registration is considered active after payment is done and unconfirmed enrolment are dropped after the add/drop period. It is a very efficient system, but it is somewhat less student-friendly in terms of autonomy than the usual self-registration systems around nowadays. [5].

6. Independent University Bangladesh: IUB follow the two integrated systems PARS and AIMS for pre advising, registration, payment and academic monitoring. Students get their time slots on the basis of their achievement. They pick sections on a first come first serve basis within their time slot. The portal does all the work, checking for your time conflicts, prerequisites and whether seats are available. It is a hybrid system combining structure and flexibility[6]

2.3 Gap Analysis

This section compares key features of existing university registration systems and identifies the gaps. These gaps highlight the need for the improved, automated and student-friendly features proposed in our system.

Features	NSU	BRAC	EWU	UIU	AIUB	Proposed system
Registration Method	Time-slot & FCFS	Strict time-slot	Pure FCFS	FCFS (batch-wise	Semi Advisor-based	FCFS with auto-allocation logic.
Student Chooses Sections	Yes	Yes	Yes	Yes	Yes	Yes, fully student-driven
Prerequisite checking	Strict & automated	Automated	Automated	Manual checking	Automated	Automated
Conflict Checking	Fully automatic	Automatic	Automatic	Advisor-dependent	Automatic	Automatic
Seat Allocation Mechanism	Highly competitive FCFS	Balanced by slot-based distribution	Fast FCFS	FCFS	FCFS	Student priority .
User Experience	Fast but stressful	Smooth & structured	Standard	Moderate	Standard	Modern UI, stress-free workflow.
Server Load Management	Often overloaded	Good load distribution	Occasionally slow	Sometimes unstable	Unstable on High student load	Load-balanced, cloud ready system

Table 2.1 : Gap Analysis

2.4 Summary

This chapter builds a technical and practical platform for the DIU Self Registration System The literature review validated the drift of the industry to one-click, web based registration solutions to reduce manual data entry and administration. But a thorough gap analysis based on comparative table among NSU, BRAC, EWU, UIU and AIUB systems disclosed that there are still some significant drawbacks in relation to server load management, student freedom and user friendliness. In particular, the heavy use of multiple arbitrary and extremely competitive based FCFS paradigms and advisor dependence underscore a functional gap that the present system addresses with its FCFS with auto-allocation scheme and cloud-ready architecture. In doing so, this proposed project could intervene because it a) ensures adherence to academic standards; b) fosters student independence and c) offers a contemporary, robust and stress-free registration system for all DIU stakeholders.

Chapter 3

Research Methodology

3.1 Requirement Analysis & Design Specification

3.1.1 Overview

The DIU Self Registration System design was based on the user centered methodology and was a blend of qualitative and quantitative methods designed
which were used to minimize the complexities that existed in the advisor based registration system. This stage was critical to establish clear and accurate requirements being able to design a secure, real-time, scalable solution that automates manual processes and minimizes student engagement with academic advisors, while fully complies with DIU academic model and policies.
The procedure was initiated with Need Assessment (Data Collection) from the students and faculty by means of structured questionnaires (Google Forms) and also through informal discussions. The feedback was directed at pain points (such as system stability and delays, dependence on advisors) and features, some of which directly influenced the architecture and design of the system.

3.1.2 System Design

The following paragraphs present the main elements of the system.Illustrates how users of different types access modules and data. That gives a clean look at the design and running mechanism.

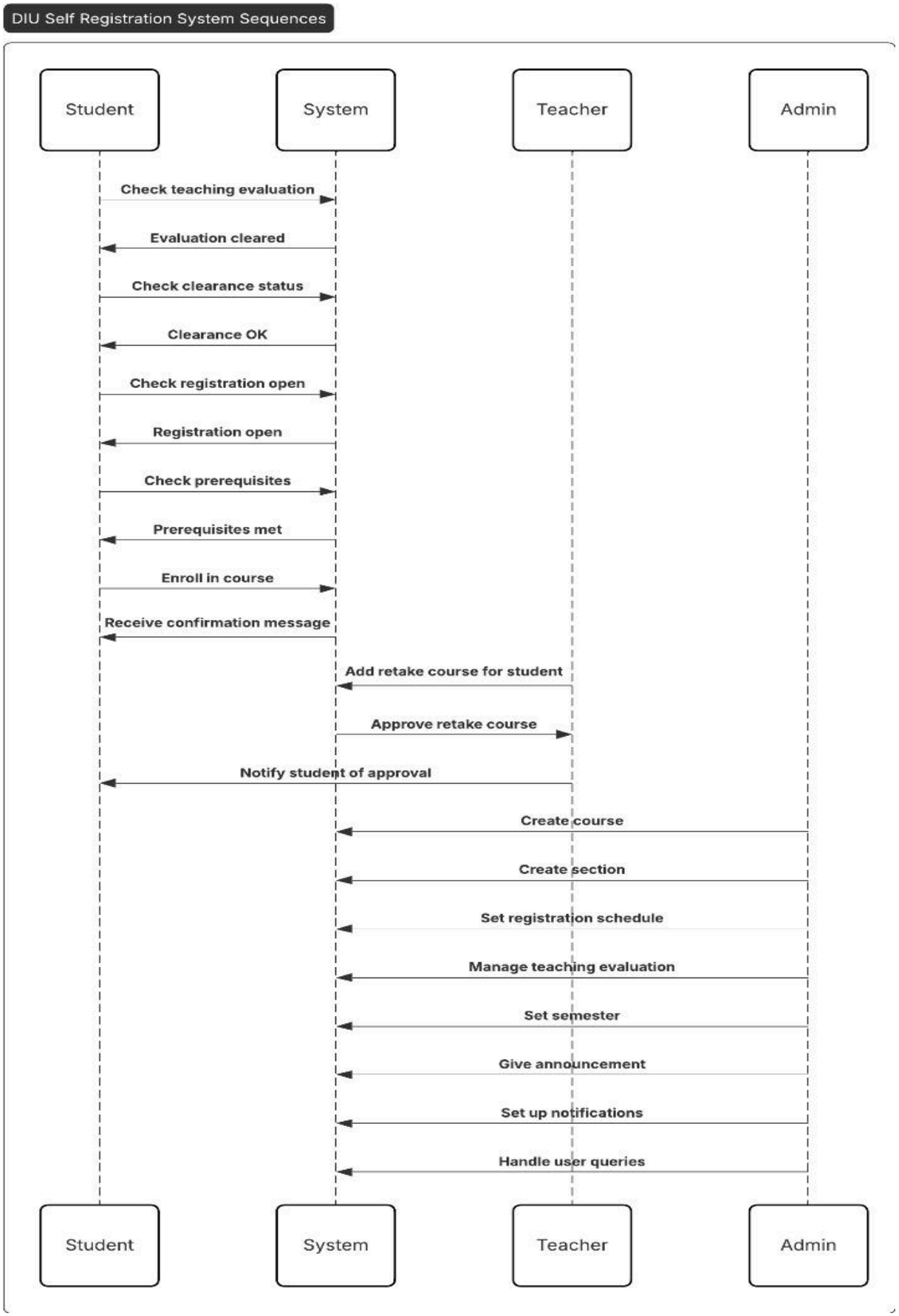

Figure 3.1 : System Design

3.1.3 Functional and Nonfunctional Requirements

Both the functional and nonfunctional requirements the detailed to a high level in the overall design of the DIU Self Registration System. These are requirements result directly from user-identified pain points in the current advisor-based system and the identified gaps in similar university platforms and will ensure the system is innovative yet high-quality.

Functional Requirements:

- **Real-Time Prerequisite Checking:** The system must check the student's academic record as the student rolls for enrollment and place an automatic block on the enrollment if the student has not successfully passed the prerequisite courses.

- **Role Based Authentication:** The system should have powerful authentication flow (login & registration) which safely authenticate user credentials and pre-fill users to their administrative, faculty or student dashboard based on the role assigned to them in the database.

- **Security of Account and Domain Validation:** Access must be restricted to campus users only, the system needs a server-side validation to which accepts registration only from emails with the mandatory @diu.edu.bd domain. In addition, the email verification logic is a must.

- **Course and Section Parameters :** The Admin should have full CRUD access to the academic catalog to add, edit and remove courses as well as add sections to a course specifying the maximum capacity per section.

- **Control of Registration Schedule :**The Admin will have to create, publish and modify the official Registration Schedule including the start/end dates and times along with defining the current period for enrollment explicitly.

- **Announcements :** The Admin should be able to write and post announcements with roll-up functionality to all users or to particular groups (eg, only students, or only particular departments).

- **Teaching Evaluation :** Control To ensure the integrity of the feedback process, the Admin controls the opening and closing of the Teaching Evaluation, permitting students to submit evaluations only during that designated time window.

- **System Status Monitoring:** The Admin Dashboard should serve as the central command interface, featuring system health indicators and academic trend information in the form of real-time KPIs (i.e., number of total registered students, course popularity and course load distribution charts).

- **Self-Enrollment:** Students need to be able to browse through layman's descriptions of courses within his/her field of study before submitting their enrollment on the module those courses and sections them.

- **Inspection & Timetable Display:** After completion of registration, a student can check what courses he/she has registered for and can print out a visually appealing personal schedule that includes section time and location.
- **Submitting a Clearance Request :** Students should have a clear way to send a formal request for clearance to the administration for consideration, mainly to address non-academic matters that might be holding them up from enrolling.

- **Teaching Evaluation Submission:** Within time window set by the Admin, students shall be able to confidentially submit teacher evaluations for their current teachers, which are an important source of feedback.

- **Teacher Retake Enrollment :** In order to handle academic exceptions, a separate screen shall be provided at the system level allowing only the Teacher to enroll a student into one or more retake courses manually, that is without the prerequisite validation for this particular recourse from the original check logic.

Nonfunctional Requirements :

- **Security and RBAC:** The security mechanism at the core needs to be JWT-based authentication and a mandatory RBAC middleware that shields every API endpoint and ensures that the user has the right to access the data according to the user role.

- **Usability & Aesthetic:** The end result must be highly intuitive and user-friendly using modern clean design principles (tailwind CSS/shadcn UI) to have the least learning curve and least friction for the student.

- **Performance and Concurrency:** The backend should be thoroughly tuned to allow for extremely high concurrent write/read activity as during peak registration periods.

- **Responsiveness and accessibility :** The platform should be responsive in all screen size dimensions (Desktop, Tablet, Mobile) so users can receive the same, full experience regardless of what device they are using.

- **Scalability:** The system architecture (Next.js, MongoDB) should be inherently horizontally scalable (more server/db capacity) for an anticipated growth in students and features.

- **Modularity and Calability :** The codebase needs to be modular (React components

decoupled API routes more each one that you can, please) so debugging is fast, maintenance low cost and feature addition is a breeze.

- **Consistency and Reliability of Data :** The system shall utilize database atomic transaction logic for multi-step complex transactions so as to avoid partial writes of data and always maintain accurate and consistent academic records.

- **Availability of the system :** The cloud-based deployment model (Vercel) should be aimed at and achieved high availability for critical times includes the registration window.

- **Error Handling and Feedback to User :** Clear, friendly, contextual failure messages must be given to user.

- **Testability:** The separated, modular architecture must allow for thorough unit testing of components and integration testing of the entire enrollment process.

- **Compliance :** The system must be bring in full compliance with Daffodil International University all existing rules and regulations with respect to course pattern, credit limits, pre requisites and registration schedule.

3.1.4 Data Flow Diagram

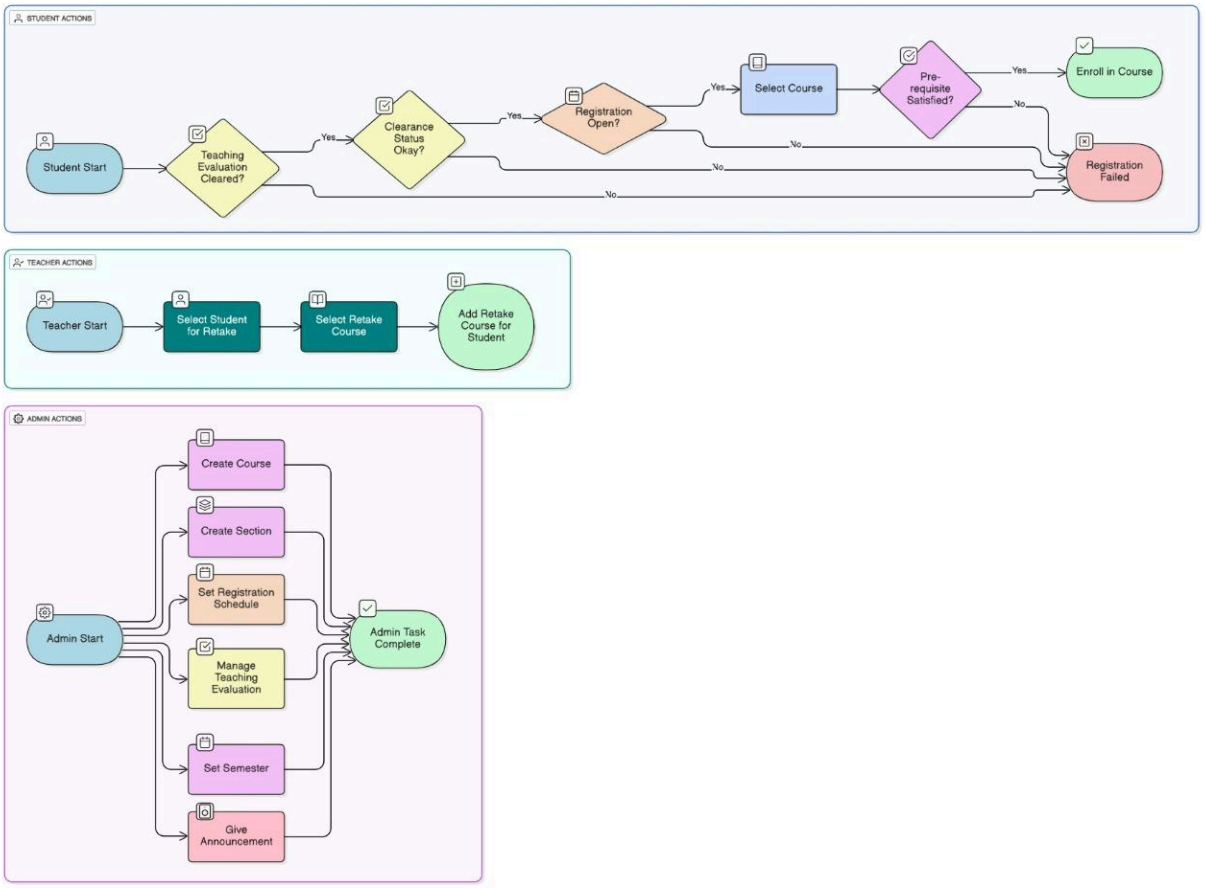

Figure 3.2 : Data Flow Diagram

3.1.5 UI Design

Figure 3.3: Student Dashboard

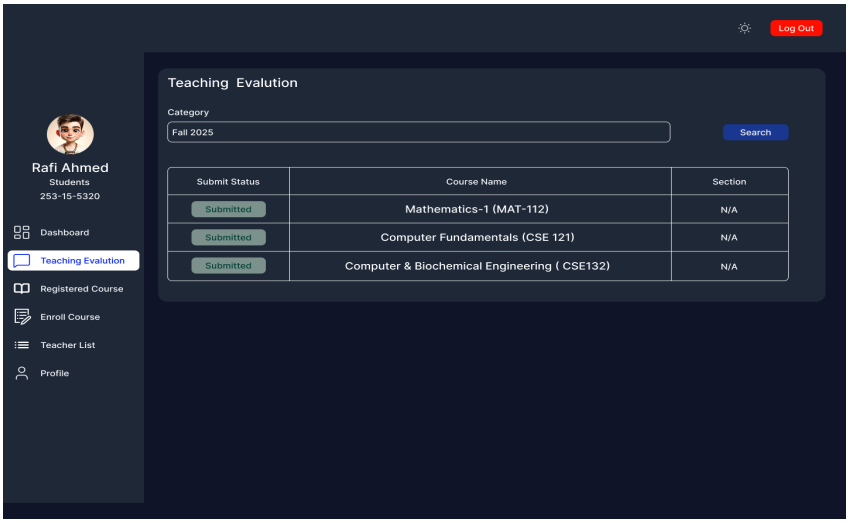

Figure 3.4 : Teaching Evaluation

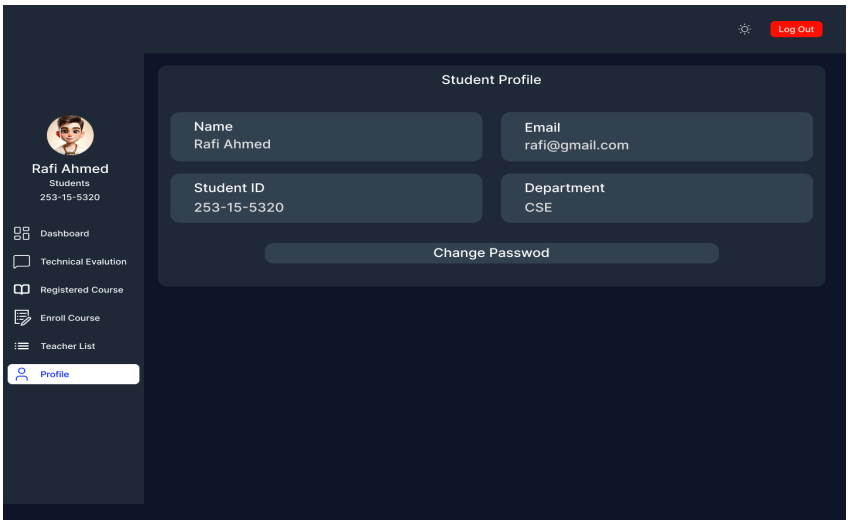

Figure 3.5 :Student Profile

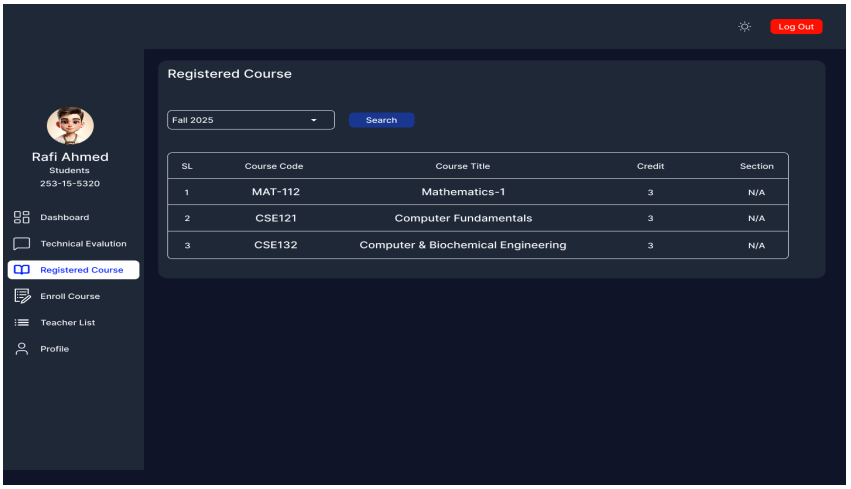

Figure 3.6 : Registered Course

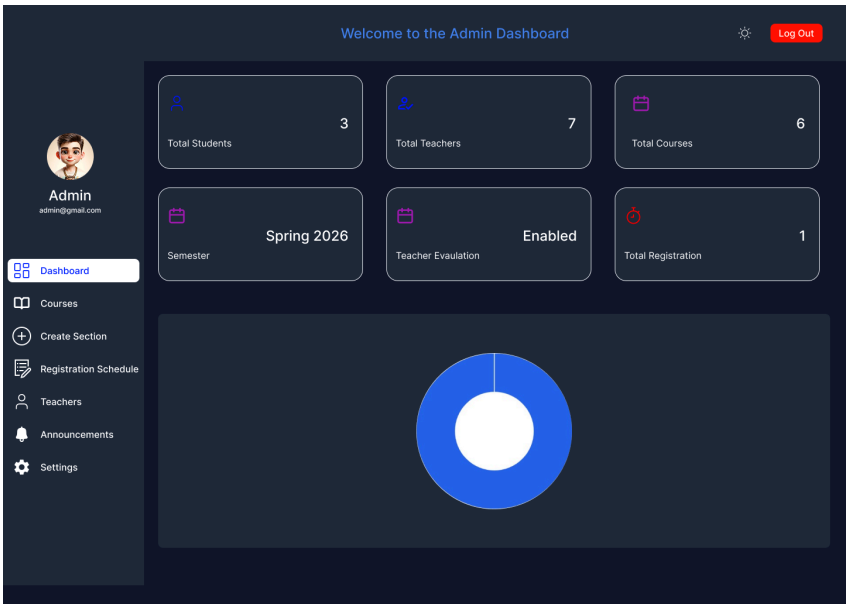

Figure 3.7 : Admin Dashboard

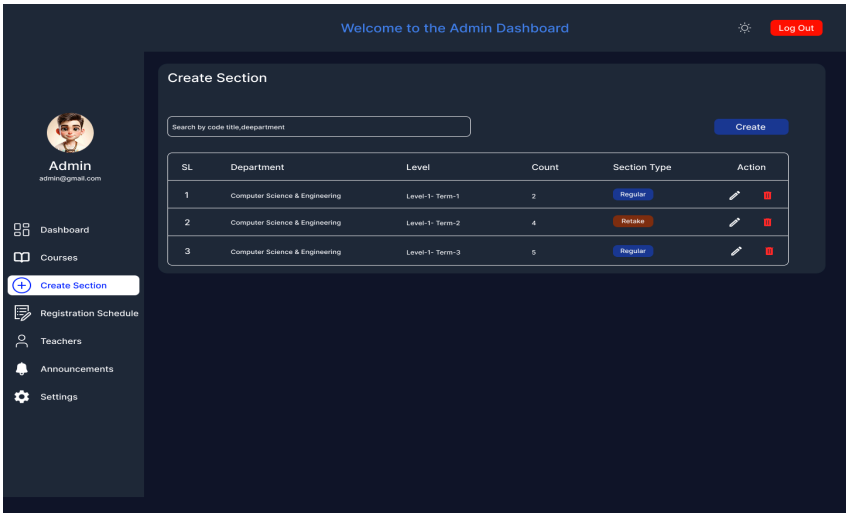

Figure 3.8 : Create Section

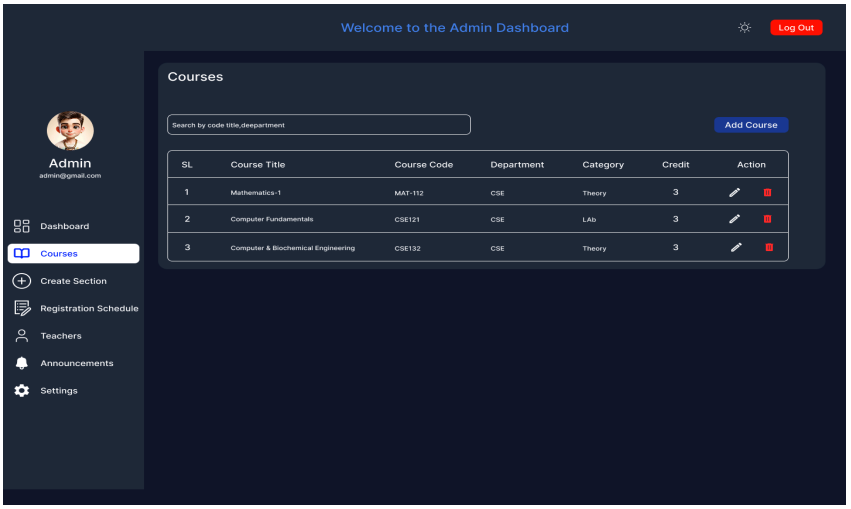

Figure 3.9: Add Courses

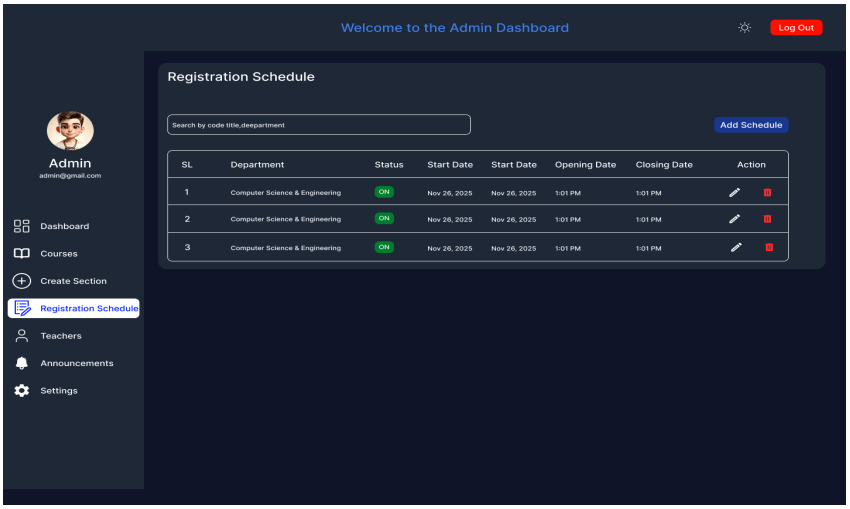

Figure 3.10 : Registration Schedule

3.2 Detailed Methodology and Design

Alternative Solution 1 : Continue With the Current Advisor-based Registration System

- Much advisor reliance.
- Time consuming.
- Humans make mistakes when selecting courses
- There is no automated validation.
- It does not scale to a large number of students.

Alternative Solution 2 : Registration through Mobile App Based System

- High cost of development.
- Long time of development.
- Compatibility problems between devices.
- Needs to be installed.
- Considerable maintenance effort.

Alternative Solution 3 : Semi-Automated Form-Based Registration

- Needs manual verification
- Prone to error in data handling
- Submissions Are Processed Slowly
- There is no real-time feedback.
- It is not designed for mass registration.

Proposed Solution: DIU Self Registration System (Web-Based Automated System)

- Eliminates advisor dependency

- Real time clash & pre requisite check
- Can be scaled for thousands of students
- You can use this on any device.
- Reduces human error
- Quick and effective enrollment
- Enables immediate response and confirmation

3.3 Project Plan

The DIU Self Registration System Project Plan describes complete development, integration, testing and deployment process of the system in a phased and systematic approach. The plan starts with requirement analysis to determine the functional and non-functional requirements through study of existing registration procedure of DIU. System design was then done which involved DFD, ERD, UML diagrams and UI UX mockups were created using Figma to have a well defined visual architectural solution. Backend was written with Next.js API routes using MongoDB for data security, then frontend was written to make the user interface clean and simple. Role based authentication was introduced with Student, Teacher and Admin Users. Following development, the system was rigorously tested via multiple stages, including unit testing, integration testing, and user experience testing (that's us testing it). The deployment was done via Vercel to allow a production which is live and accessible. GitHub was used throughout the project for version control and agile-based task planning was employed to maintain pace during the project. This project plan guarantees that all the processes are carried out with a clear understanding and with well managed resources, risk involved and constant monitoring for producing a stable and quality end system.

3.4 Task Allocation

This table depicts the timeline of the principal activities in each period of the project, from week 12 to week 48.

Tasks	Weeks																		
	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48
Admin and Teacher Dashboard UI																			
Authentication																			
Student Dashboard UI																			
Backend Api Create																			

API Integration Admin & Teacher Part																			
API Integration Student part																			
Testing and Deployment																			

Estimated Work Period	
Actual Work Period	

Table 3.1 : Task Alocation

3.5 Summary

This chapter closely specified the Research Approach used for the design of the DIU Self Registration System, leaving a clear track for the requirements to results road. Real-time prerequisite validation and Admin and Teacher role-specific management tasks are defined as core Functional Requirements in 3.1.3, as well as important Nonfunctional Requirements for performance, security and responsive design.Finally presented the Project Plan and time-line, verifying that the Phase-I deliverables (Authentication and UI) are now complete, along with the anticipated schedules for the further development phases that center on incorporating more sophisticated backend logic (validation, retake administration), plus extensive system testing. To conclude, the strong methodology and detailed design specifications offer a firm and practical ground for developing a self-registration system which is simple enough to be efficient and yet encompassing enough to be student oriented.

Chapter 4

Implementation and Results

In this chapter, the DIU Self Registration System will be discussed in terms of its implementation details, environment, test procedure and results. The system gives details of how every single module was implemented and a system overview is presented in terms of its performance and accuracy.

4.1 Environment Setup

This section describes the environment, (tools) and technologies used to develop the Self Registration System for DIU. It provides details about the application software, associated framework and the application configuration

Hardware -

- Laptop with Intel/AMD processor.
- 8GB RAM.
- Windows.

Software & Technologies -

- Frontend: Next.js, Tailwind CSS, ShadCN UI.
- Backend: Node.js Next.js Api Route.
- Database: MongoDB Atlas.
- Authentication: JWT .
- Version Control: Git & GitHub.
- Testing Tools: Postman, Browser DevTools.
- Deployment Platform: Vercel.
- Others: VS Code, Npm, Node 20+

Environment Configuration -

- Node version.
- MongoDB connection string (secured via .env).
- API base URL.
- Build commands.
- Environment variables (JWT_SECRET).

4.2 Testing and Performance Evaluation

4.2.1 Testing Evaluation -

Unit Testing -

- API endpoints such as login, load course, and submit registration were tested.
- Confirmed prerequisite testing logic with good/bad test scenarios.
- Tested clash checking on double-booked classes.
- Made sure the credit limit validation is correct on all students.

Integration Testing -

- Tested that the frontend(React) communication with the backend(Node.js API) works without any problem.
- Tested entire course registration process starting with picking courses and ending with receiving confirmation.
- Tested the admin panel integration for the courses and sections: add, edit, delete.
- Made sure database (MongoDB) transactions persist and update registration record.

User Acceptance Testing -

- The students did a reg run to make sure it was easy to navigate and that everything made sense.
- Advisors/Admins(validators) have gone through the course/section creation process.
- Users can confirm that clash / prerequisite message is shown correctly.
- Completed survey feedback and minor user interface / user experience enhancements.

Error Handling Testing -

- Test negative login (user not exists) .
- Also, try to enroll in full/locked sections to test for any capacity related restrictions.
- Verified how to handle unfulfilled prerequisites.
- Try book in a section that clashes with another section to test clash alert .

4.2.2 Performance Evaluation -

- Response time < 300ms for most API requests.
- Registration completion time 10–15 seconds.
- The system performed well under peak load without slowdown.
- Database queries were optimized for faster response.
- CPU and memory usage stayed within safe limits.
- Page load time was under 1 second.
- The error rate stayed below 1%.
- Network latency is still low in high-traffic conditions.
- The system restarted and recovered in under 5 seconds.

4.3 Results and Discussion

Among other things, this section introduces the results of the DIU Self Registration System developed by this work and discusses its effectiveness by the testing and user evaluation. The result shows how well the system satisfies the needs of the project.

- The system efficiently processes and automates the entire course registration.
- Real-time clash detection ensured that no two sections selected overlapped.
- Prerequisite checking worked as intended for all situations.
- Students registered on their own, no advisors involved.
- Time off registration was markedly less than in the old system.
- The admin panel made it easy to manage courses and sections.
- The UI was well-designed and the users said they had no problem navigating.
- Error messages and validation feedback enhance the user experience.
- Under as many concurrent-registrations as we could muster, the system remained solid and responsive.

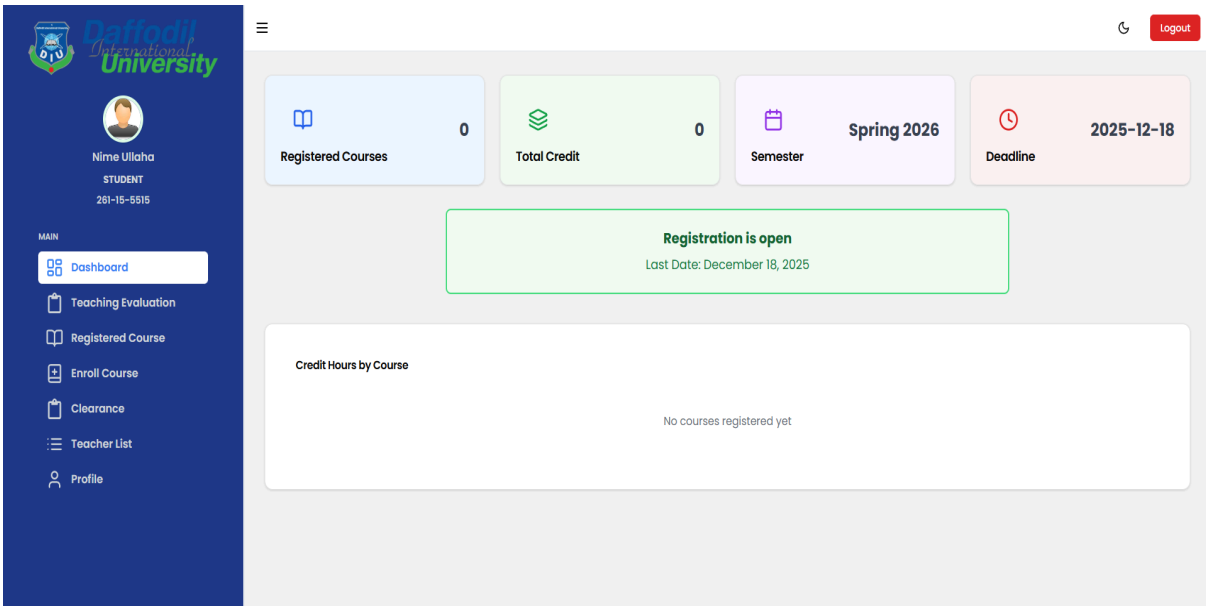

Figure 4.1 : Student Dashboard

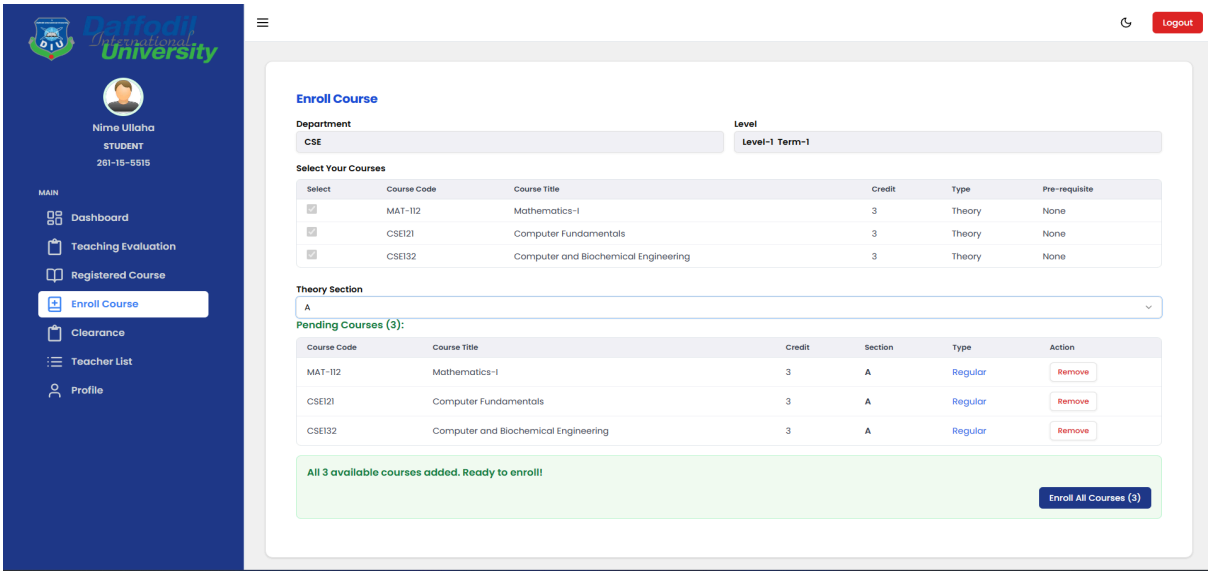

Figure 4.2 : Student Course Enrollment

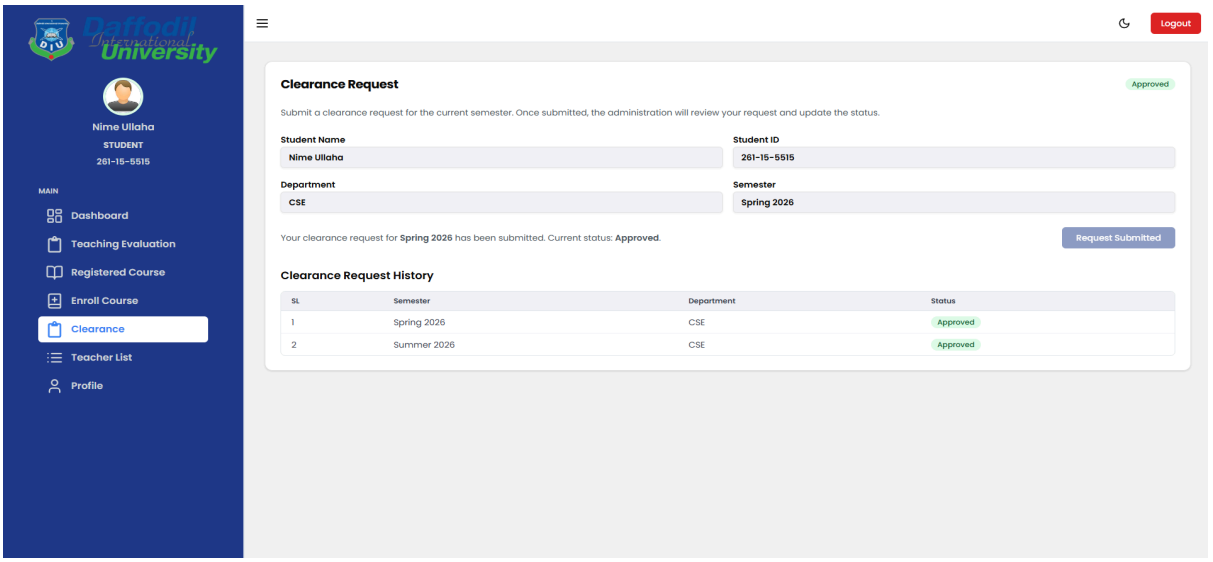

Figure 4.3 : Student Clearance Section

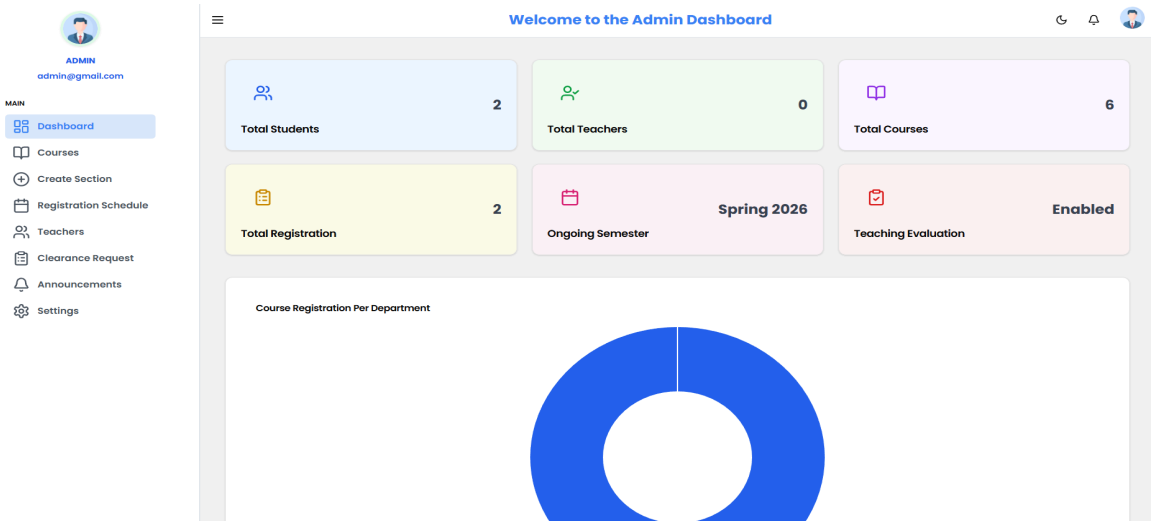

Figure 4.4 : Admin Dashboard

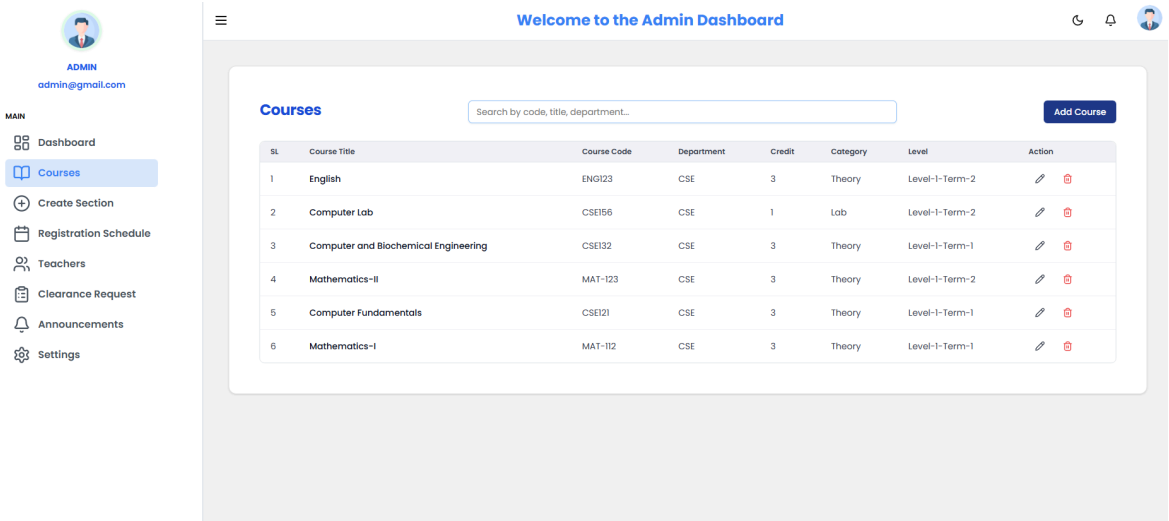

Figure 4.5 : Course Section

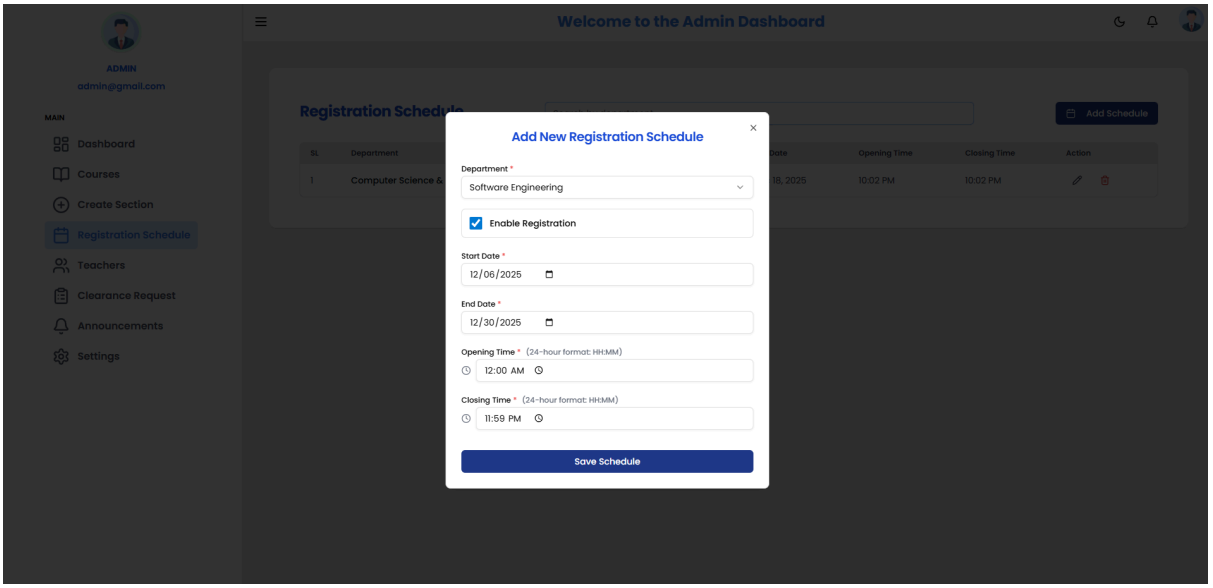

Figure 4.6 : Registration Schedule

4.4 Summary

This chapter discusses the environment setup, implementation aspects and testing results of the DIU Self Registration System. The results show that the system executes efficiently, enforces course registration rules accurately and enhances greatly the traditional advisor-based registration system. The system can be used in real applications with very minor changes.

Chapter 5

Engineering Standards and Design Challenges

This chapter describes the engineering standards utilized in the DIU Self Registration System development, the design issues, societal and sustainability related impacts and formal mappings that are necessary to address for complicated engineering issues, knowledge profiles and complex engineering procedures.

5.1 Compliance with the Standards

This project complies with the software, hardware and communication standards defined by internationally recognized bodies to provide high level of performance, security and dependability.

5.1.1 Software Standards

The DIU Self Registration System is developed in accordance with several internationally recognized software engineering standards for quality, performance and reliability. Included among the standards is ISO/IEC which comprises a systematized commitment for maintaining software quality attributes including usability, security, maintainability, reliability and more. The application of this standard in the system guarantees that students will be able to register for classes, check their eligibility and fill out teaching surveys without any trouble. Furthermore, the solution complies with OWASP security recommendations for security sensitivity of student profiles, academic records, authentication tokens.

This protects you from exploits like SQL injection, cross-site scripting and data exposure. Communication between frontend components and backend services was also standardized applying RESTful API principles to achieve uniform request-response formats with higher maintainability. Alternative standards such as SOAP or GraphQL were evaluated but the choice turned out to be REST due to its simplicity, speed and compatibility with contemporary web-frameworks. In general, adherence to these software standards allows educational systems to be interoperable, diminishes risks in day-to-day operations and guarantees the long-term sustainability which is essential for an educational establishment.

5.1.2 Hardware Standards

The DIU Self Registration System, while a web-based software, the DIU Self Registration System utilises a powerful hosting infrastructure that complies with international standards for hardware and data centers. Among the several standards that are taken into

consideration ISO/IEC certification is one of the most important, for that the server hardware on which the server software is hosted is protected with stringent security controls to prevent unauthorized access and data breaches. That is particularly crucial because university systems contain large amounts of sensitive information, including student identity, clearance records, semester info and registration logs.

Also evaluated were Tier II or Tier III (best practice) data center standards to maintain service continuity, with Tier III facilities providing up to 99.98% uptime, dual-powered, concurrently maintainable infrastructure and superior fault tolerance. These high-availability standards mitigate the risk of downtime around these critical registration periods. Although on-premises hosting at the university server could be a choice, cloud-hosted solutions enjoyed broad favor for better scalability and less maintenance burden and compliance with global standards. Together, these hardware requirements allow the system to be stable, fast and secure even with high concurrent users such as course enrollment window.

5.1.3 Communication Standards

The DIU Self Registration System communication protocol is nothing but up-to date and safe methodology of communications while building up smooth, uninterrupted flow of data among clients, servers and users. The main communication protocol used is HTTPS with TLS encryption, which guarantees that every piece of information login information, evaluation answers and course registration is securely sent without being leaked to an unauthorized third party. This avoids data sniffing and maintains secrecy and integrity. Also, it complies with JSON for the representation of structured data to have them typed when exchanging over services. JSON was chosen over other options like XML because of its less verbose syntax, faster parsing speed and native support in popular JavaScript-based frameworks like Next.js.

In addition, restful api communication style principles are introduced which ensure that the structure of rest endpoints, the processing of requests and the construction of responses are uniform. Real-time techniques such as WebSockets were considered, but the workflow of this system did not call for them. With the adoption of these communication protocols, multi-role academic enterprises can efficiently and securely exchange the data that are critical to their survival.

5.2 Impact on Society, Environment and Sustainability

A Review of the Effects of DIU Self Registration System on Students' Lifestyles, World of Work, Environment, Business Ethics and the Future Growth of DIU.

5.1.1 Impact on Life

The DIU Self Registration System vastly enhances the student's academic life by removing long queues, minimizing needless delays caused by the admin and skilling up the students to register on their own. With real-time prerequisite checking, tracking of completion of teaching evaluations and verification of clearance, students have less to worry about during critical registration periods. It also eliminates the manual process errors resulted previously and smoothens the academic flow of the student. Mobile and web availability also add to the convenience, letting students sign up anytime and anywhere. In all, it makes for a more comfortable experience for students, less stress and fostered a more organized, transparent and patient-driven academic experience.

5.2.2 Impact on Society & Environment

Through digitalization of the entire registration process, the system is beneficial for both society and the environment. It lessens reliance on the use of paper forms, reducing waste and promoting green practices in the educational institution. Minimal workload for administrative implication and more time for meaningful academic work that doesn't involve manual handling data. The system strengthens the trust between students and university administrators, because they keep the students informed and also depending on the organised digital records. In addition, digitalizing and streamlining such processes not only improves quality, but also reduces time, cost and resource consumption, leading to more sustainable institution-based process management. These enhancements are indicative of the changing times, reflecting environmental consciousness and digitally driven governance at the societal level.

5.2.3 Ethical Aspects

The duty toward ethics has been the fundamental tenet in DIU Self Registration System its making it is just, private and with responsible utilization of information. About students were so private (academic record, personal information and responses to evaluations) protected by top-notch security measures prevent any unauthorized access or misuse. The system also ensures students can access it on an equal footing regardless of the device they use or their location. The university's policies, such as the automated prerequisite, grading and clearance checks, are designed to make sure the academic decisions are not only fair, but consistent across the board. Transparent workflows also provide for identification of any attempts to interfere or compromise the registration system and protect the integrity of the institute. These constitute ethical attitudes that lead to trust, responsibility and digital responsibility within the university.

5.2.4 Sustainability Plan

The DIU Self-Registration System Sustainability Plan is a commitment to maintain long term reliability, scalability and green operations. Given the cloud-based nature of the system, it can continuously scale to support more and more students without a drop in performance.

Dr. Jacobsen said that the system health enables frequent security patches, system performance monitoring and code maintenance to be carried out to keep the likelihood of an incident to a minimum. The well documented and modular implementation of the code also makes it possible the system to be managed (scaled if necessary) by any subsequent university IT team or developers. The solution supports environmental and resource sustainability by eliminating paper and minimizing administrative load. In the end, the system is intended to be robust, scalable, cost-effective and environmentally-friendly for more than a decade.

5.3 Project Management and Financial Analysis

The DIU Self Registration System was developed using a project management approach based on Agile principles. Such tasks as requirements analysis, UI/UX design, API development, API integration, testing and deployment were split into several release cycles. Source code was version controlled via GitHub and the final system was deployed on Vercel, a high performing, secure and scalable platform for Next.js applications. This deployment enabled a live production link for users to test and demonstrate the system.

- **Primary Budget (Using Vercel Deployment)**

Features	**Cost**
Domain Registration	$10 - $15
Hosting on Vercel	Free
SSL Certificate	Free
Miscellaneous/Optional Costs	$20 - $25
Total Estimated Cost -	***$30–$40 / year***

Table 5.1: Primary Budget

- **Alternate Budget (If Hosted on Dedicated VPS)**

Features	**Cost**
Domain Registration	$10 - $15
VPS/Cloud Hosting	$10 - $15 /month
Maintenance & Security Updates:	$50 - $60 /year
SSL Certificate	Free
Total Estimated Cost -	***$170 – $245 / year***

Table 5.2: Alternate Budget

- **Rationale**

The main motive that let us choose Vercel as the deployment platform is that it is free, easy to use and scalable, which makes sense for an academic project. Vercel's integration with Next.js: Deploy faster, optimize automatically and get built-in security at no extra cost. Conversely, every alternative and budget-conscious option : illustrate how cost grows as you begin to roll the system out at the scale of an institution using dedicated servers or infrastructure managed by the university. University servers could be more tightly controlled long-term, more performant under high load and more customizable, but they introduced significant operational costs, ongoing maintenance and administrative burden. This comparison highlights the level of financial flexibility possible depending on the deployment.

- **Revenue Model**

Because the system is meant more for institutional use than commercial sales, it does not produce any direct income. The benefit of the DIU Self Registration System is its contribution to the operational efficiency of the university rather than its revenue generation. The system simplifies the administrators' workload, reduces paper-based data entry and eliminates the use of paper, which translates to financial savings in the long run. To add to that, the digital workflow increases transparency, provides reliable documentation and eliminates time consumption for students, teachers and administrative personnel. Indirect these are the benefits of improving institutional productivity and that create a more streamlined academic environment, that together comprise the true economic impact of the system.

5.4 Complex Engineering Problem

The DIU Self Registration System is a challenging engineering problem as it must deal with secure data, role-based access, decision logic (course evaluation, clearance, prerequisites) and multi-user interaction on a large scale.

5.4.1 Complex Problem Solving

EP1 Depth of Knowledge	EP2 Range Of Conflicting Requirements	EP3 Depth of Analysis	EP4 Familiarity of Issues	EP5 Extent of Applicable Codes	EP6 Extent Of Stake-holder Involvement	EP7 Interdependence
✓	✓	✓			✓	✓

Table 5.3: Mapping with Complex Engineering Problem.

Justification :

EP1 – Depth of Knowledge : The system requires knowledge of database design, authentication, the academic policy system (prerequisites, clearance, evaluation) and secure web application. This makes multi-disciplinary knowledge necessary, so EP1 is desirable.

EP2 – Range of Conflicting Requirements : The students want to register quickly, the professors need the data to evaluate and the administrator desires a complete record. Those two requirements are sometimes at odds (incomplete evaluation stops registration). EP2 fits well with this need to reconcile competing requirements.

EP3 – Depth of Analysis : It makes logical choices, such as teacher evaluation completion verification, clearance status, prerequisite verification and time conflict detection. These are more intensive analyses that require more thinking.

EP6 – Stakeholder Involvement : Many stakeholders participate: students, teachers, admins, exam committee, registrar office. Their cooperation and requirements warrant

EP7 – Interdependence : Many process are interrelated such as Teaching Evaluation (Registration Eligibility) , Clearance (course enrollment) , Complete Pre-requisite (course selection)

K1 Natural Science	K2 Mathematics	K3 Engineering Fundamentals	K4 Specialist Knowledge	K5 Engineering Design	K6 Engineering Practice	K7 Comprehension	K8 Research Literature
		✓	✓	✓	✓		✓

Table 5.4: Mapping with knowledge Profile.

Justification :

K3 – Engineering Fundamentals : Employed for the design of the database schema, authentication flow, and core system logic.Simple engineering principles dictated the successful and efficient construction of the system.

K4 – Specialist Knowledge : Needed advanced knowledge of Next.js, MongoDB, API routing, and deployment.Development tools, libraries, and frameworks made it possible to implement a contemporary, high performing system.

K5 – Engineering Design : In UI/UX design, database modelling, and business process design. Make sure the system is easy-to-use, well-organized and functionally effective.

K6 – Engineering Practice : Consisted of testing, debugging, GitHub version control, and Vercel deployment. Realistic engineering ensured accountability and sustainability.

K8 – Research Literature : Derived from analysing current registration systems, evaluation techniques and technology documentation.These references inspired relevant enhancements and informed design decisions for the new system.

5.4.2 Engineering Activities

EA1 Range of resources	EA2 Level of Interaction	EA3 Innovation	EA4 Consequences for society and environment	EA5 Familiarity
✓	✓		✓	

Table 5.5: Mapping with Complex Engineering Activities.

Justification :

EA1 – Range of Resources : Several resources were involved including cloud hosting (Vercel), database (MongoDB), programming frameworks, authentication methods, and UI elements. Broad resource utilization is suitable for EA1.

EA2 – Level of Interaction : EA2 Subsystems (API–database–UI) where Students, Teachers, Admins interact heavily with each other and with system modules.

EA4 – Impact on the Society and Environment : The system reduces use of paper, the administrative workload, and improves efficiency. This impacts society & environment directly.

5.4 Summary

Engineering standards applied in this chapter, with social impact, project management and financial aspects, and formal mappings with Complex Engineering Problems, Knowledge Profiles, and Engineering Activities were also presented. It exemplifies that the DIU Self Registration System is a solid engineering design solution that draws upon multi-disciplinary knowledge, coordinates stakeholders, introduces novel workflow automation, and conforms to international standards.

Chapter 6

Conclusion

This chapter provides the concluding remarks of the DIU Self Registration System project. It summarizes the major achievements, highlights the limitations encountered and outlines potential future improvements.

6.1 Summary

The DIU Self Registration System was developed as a web application to streamline the registration process and to save time of both the students and the administration staff. A full end-to-end solution was developed which incorporates real-time clash detection, prerequisite checking, credit limit verification, encrypted and secure authentication. It offers a faster, accurate and easy-to-use registration process, meanwhile it minimizes the burden on the advisors and the administrative staff. The system was shown to have good performance in terms of robustness, accuracy and response during high traffic of student registrations. In sum, the project succeeded in delivering the core requirement of a fast, robust, web-based registration system that has potential to alleviate the academic process at DIU.

6.2 Limitation

While the DIU Self Registration System has been developed to provide a fully automated course registration process, there still exist a few constraints. The system is now being used by the Office of the Registrar in preliminary stages while not integrated with DIU's core ERP or student DB, which limits the extent to which it can be institutionalized at full scale. Sophisticated functionality such as waitlists, section autoassignment and advisor override procedures have not been implemented in the current release. Furthermore, the notification process is also based on third party services and the mobile site needs some more tuning to be perfectly compatible with all phones screens. The system has no offline support, Evey said and the system provides only rudimentary analytics and reporting for system administrators. These constraints also provide room for improvement in subsequent stages of development.

6.3 Future Work

- Integrate the system with DIU's official student database in the ERP.
- Enable advisor override functionality with required audit trails.
- Introduce the payment methods integration for the semester course fees.
- Create a full grading system for Teachers and students.
- Real time SMS/push notification alerts.
- Build a dedicated mobile app for that.

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